Intelligenza Artificiale

“Bayesian networks I”

Nicola Bellotto

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Lecture outline

- Motivation
- Product decomposition
- Conditional probability tables
- Inference by enumeration
- Examples

Slides partly based on Russell & Norvig instructors material http://aima.cs.berkeley.edu/instructors.html

Full joint distribution

- In theory, given the full joint distribution of a set of random variables, it is possible to answer **any query** by applying the rules of probability
- E.g., if we know

$$\mathbf{P}(Disease, Symptom_A, Symptom_B) \equiv \mathbf{P}(D, A, B)$$

we can answer any query, such as

$$\mathbf{P}(D|A, B) = \frac{\mathbf{P}(D, A, B)}{\mathbf{P}(A, B)} = \frac{\mathbf{P}(D, A, B)}{\sum_D \mathbf{P}(D, A, B)}$$

$$\mathbf{P}(A|D) = \frac{\mathbf{P}(D, A)}{\mathbf{P}(D)} = \frac{\sum_B \mathbf{P}(D, A, B)}{\sum_{A,B} \mathbf{P}(D, A, B)}$$

Limitations

- However, the size of the joint distribution grows exponentially with the number of variables!
- E.g., assuming 3 boolean variables, we need $2^3 = 8$ values (actually 7: the last one is implicit because the sum must be 1)

D	A	B	$P(D, A, B)$
true	*t*	*t*	0.05
t	*t*	*f*	0.08
t	*f*	*t*	0.12
t	*f*	*f*	0.06
false	*t*	*t*	0.09
f	*t*	*f*	0.04
f	*f*	*t*	0.11
f	*f*	*f*	(0.45)

Problems with full joint distributions

- Obvious problems:
  - Worst-case complexity $O(a^n)$, where a is the largest arity (e.g. $a = 2$ for boolean) and n the number of variables
  - How to find the probabilities for $O(a^n)$ entries?
  - And where to store them?

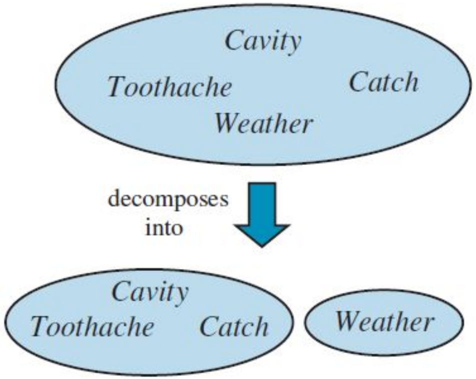
- Fortunately in many cases it is possible to reduce large distributions into smaller ones thanks to **independence**

Factorising using independence

- Recall independence

  $P(a|b) = P(a)$ or $P(b|a) = P(b)$ or $P(a, b) = P(a)P(b)$

- **Factorisation** and independence reduce large joint distributions into smaller ones



- E.g. dental problems do not influence the weather (independent)

  $\mathbf{P}(Toothache, Catch, Cavity, Weather)$ (16 values)

  $= \mathbf{P}(Weather \mid \cancel{Toothache, Catch, Cavity})\, \mathbf{P}(Toothache, Catch, Cavity)$

  $= \mathbf{P}(Weather)\, \mathbf{P}(Toothache, Catch, Cavity)$ (2+8=**10** values)

Chain rule

- To simplify, let's start by decomposing the joint distribution in a product of conditional probabilities, a procedure called **chain rule**

$$
\begin{aligned}
P(x_1, \ldots, x_n) &= P(x_n | x_{n-1}, \ldots, x_1) P(x_{n-1}, \ldots, x_1) \\
&= P(x_n | x_{n-1}, \ldots, x_1) P(x_{n-1} | x_{n-2}, \ldots, x_1) P(x_{n-2}, \ldots, x_1) \\
&= \ldots \\
&= P(x_n | x_{n-1}, \ldots, x_1) P(x_{n-1} | x_{n-2}, \ldots, x_1) \ldots P(x_2 | x_1) P(x_1) \\
&= \mathbf{\Pi}_j P(x_j | x_1, \ldots, x_{j-1})
\end{aligned}
$$

- For example

$$P(x_1, x_2, x_3) = P(x_3 | x_1, x_2) P(x_2 | x_1) P(x_1)$$

Product decomposition

- If X_j is independent from all but a subset of its predecessors (**parents**)

$$\Pi_j \mathbf{P}(X_j | X_1, \ldots, X_{j-1}) = \Pi_j \mathbf{P}(X_j | Parents(X_j))$$

where

$$Parents(X_j) \subseteq \{X_1, \ldots, X_{j-1}\}$$

then we can write the joint distribution as a product of conditional probabilities of all the variables given their parents (**product decomposition**)

$$\boxed{\mathbf{P}(X_1, \ldots, X_n) = \Pi_j \mathbf{P}(X_j | Parents(X_j))}$$

Product decomposition example

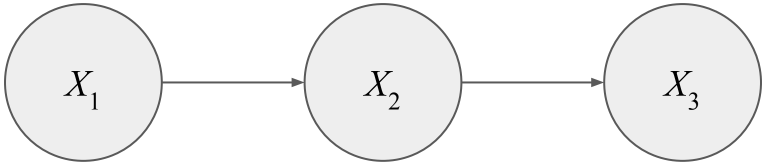
- For example

$$\mathbf{P}(X_1, X_2, X_3) = \mathbf{P}(X_3 \mid X_1, X_2)\, \mathbf{P}(X_2 \mid X_1)\, \mathbf{P}(X_1)$$

- If X_3 is (conditionally) independent of X_1 given X_2, then

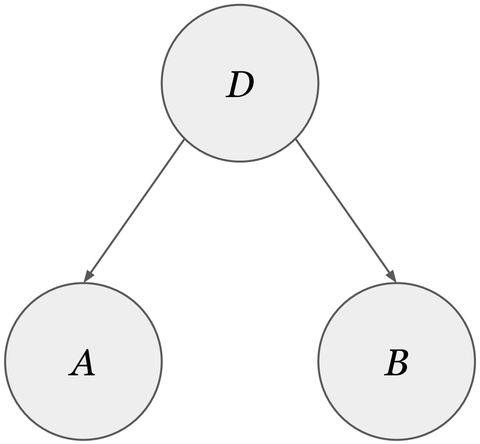
$$\mathbf{P}(X_1, X_2, X_3) = \mathbf{P}(X_3 \mid X_2)\, \mathbf{P}(X_2 \mid X_1)\, \mathbf{P}(X_1)$$

- In this case
  - X_2 is the parent of X_3
  - X_1 is the parent of X_2
  - X_1 has no parents



From full joint distributions to Bayesian networks

- In general, the joint distribution can be represented by a **Bayesian Network (BN)**
    - Also called **Belief Network**, **Probabilistic Graphical Model (PGM)**, **Directed Acyclic Graph (DAG)**



- The **nodes** are random variables and the **edges** are the dependencies between them

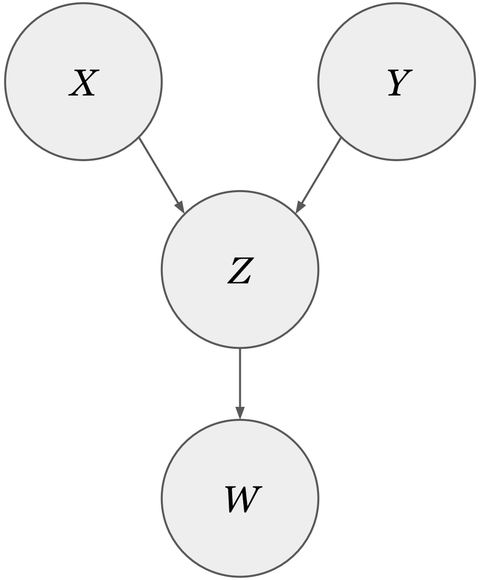
- For example

$$\mathbf{P}(A, B, D) = \mathbf{P}(A|Parents(A))\mathbf{P}(B|Parents(B))\mathbf{P}(D|Parents(D))$$
$$= \mathbf{P}(A|D)\mathbf{P}(B|D)\mathbf{P}(D)$$

Example

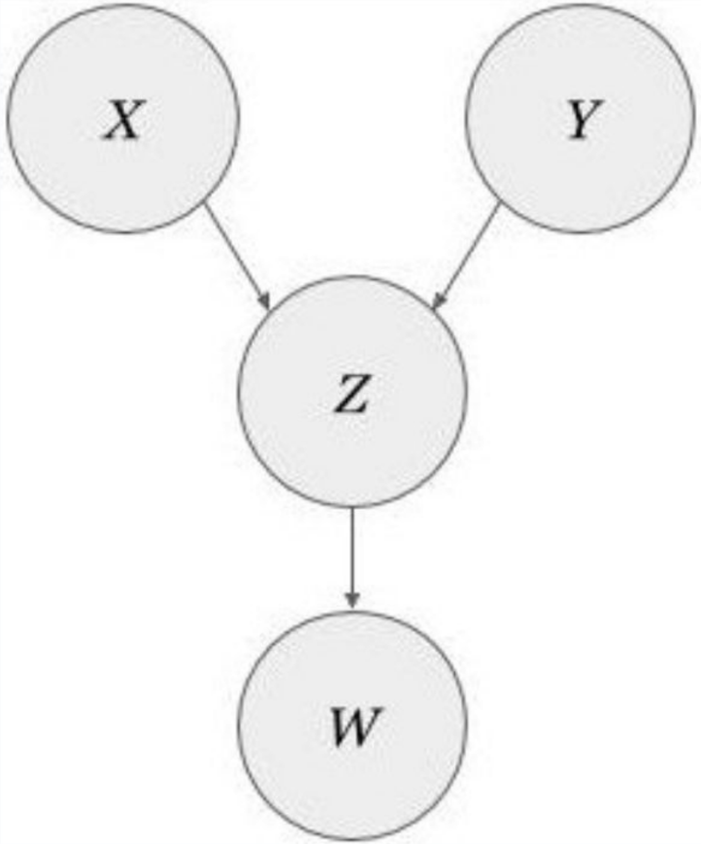
- What is the joint distribution represented by this Bayesian Network?

$$\mathbf{P}(W, Z, X, Y) = \ ?$$



Joint Distribution of Bayesian Network

0



P(W)P(Z)P(X)P(Y) 0%

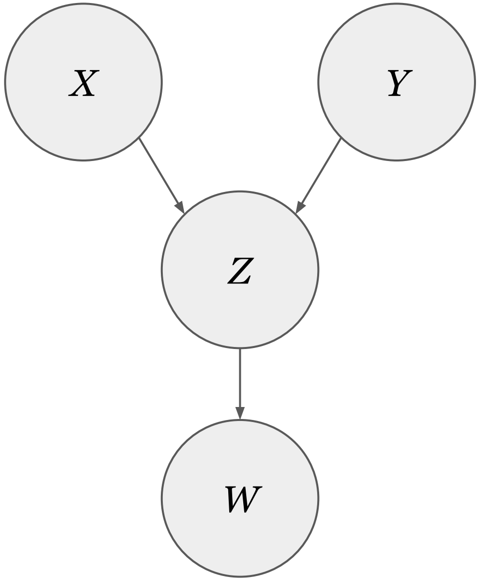
P(W|Z)P(Z|X,Y)P(X)P(Y) 0%

P(W,Z)P(Z,X)P(Z,Y) 0%

P(W|Z,X,Y)P(Z|X,Y)P(X)P(Y) 0%

None of the above 0%

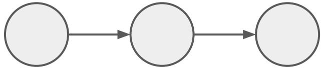
Example



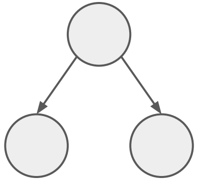
$$
\begin{aligned}
&\mathbf{P}(W, Z, X, Y) \\
&= \mathbf{P}(W|Parents(W))\mathbf{P}(Z|Parents(Z))\mathbf{P}(X|Parents(X))\mathbf{P}(Y|Parents(Y)) \\
&= \mathbf{P}(W|Z)\mathbf{P}(Z|X, Y)\mathbf{P}(X)\mathbf{P}(Y)
\end{aligned}
$$

Bayesian network structures

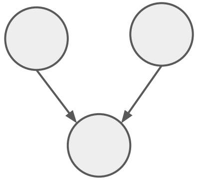
- There are three main structures
  - **Chain**



  - **Fork**



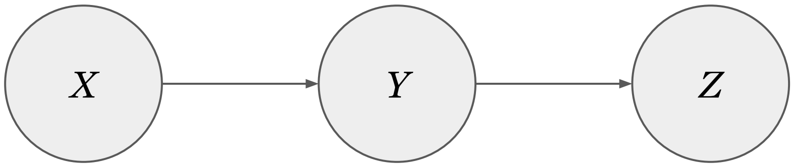
  - **Collider**



Chain

- A **chain** or **head-to-tail** connection is a case in which

$$\mathbf{P}(X, Y, Z) = \mathbf{P}(Z|Y)\mathbf{P}(Y|X)\mathbf{P}(X)$$



- The two variables X and Z are conditionally independent given Y

$$\mathbf{P}(Z|X, Y) = \frac{\mathbf{P}(Z|Y)\mathbf{P}(Y|X)\mathbf{P}(X)}{\mathbf{P}(X, Y)} = \frac{\mathbf{P}(Z|Y)\mathbf{P}(X, Y)}{\mathbf{P}(X, Y)} = \mathbf{P}(Z|Y)$$

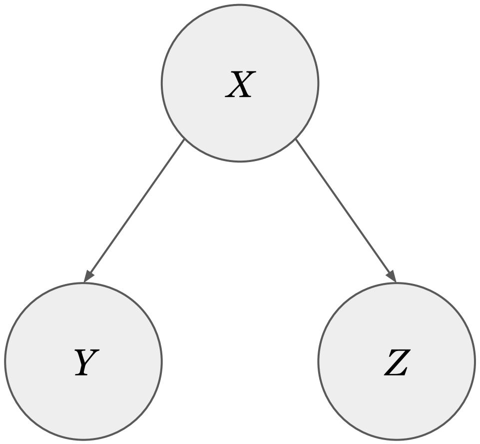
Fork

- A **fork** or **tail-to-tail** connection is a case in which

$$\mathbf{P}(X, Y, Z) = \mathbf{P}(Y|X)\mathbf{P}(Z|X)\mathbf{P}(X)$$

- The two variables Y and Z are <u>conditionally independent</u> given X

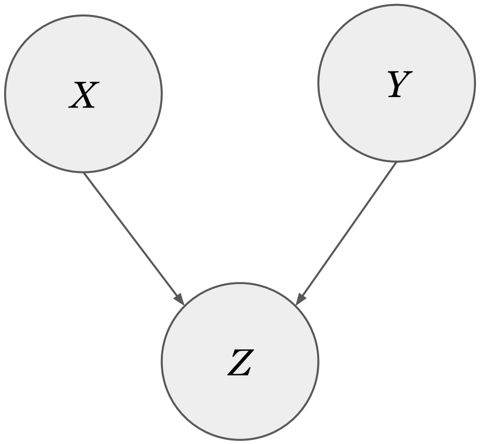
$$\mathbf{P}(Y, Z|X) = \frac{\mathbf{P}(Y|X)\mathbf{P}(Z|X)\mathbf{P}(X)}{\mathbf{P}(X)} = \mathbf{P}(Y|X)\mathbf{P}(Z|X)$$



Collider

- A **collider** or **head-to-head** connection is a case in which

$$\mathbf{P}(X, Y, Z) = \mathbf{P}(Z|X, Y)\mathbf{P}(X)\mathbf{P}(Y)$$

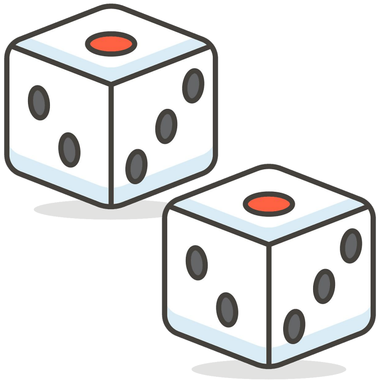


- The two variables X and Y are <u>independent</u> (**not** given Z)

$$\mathbf{P}(X, Y) = \sum_{Z} \mathbf{P}(Z|X, Y)\mathbf{P}(X)\mathbf{P}(Y) = \mathbf{P}(X)\mathbf{P}(Y)$$

Collider – example

- X is the outcome of the first die
- Y is the outcome of the second die
- Z is the sum of the two outcomes

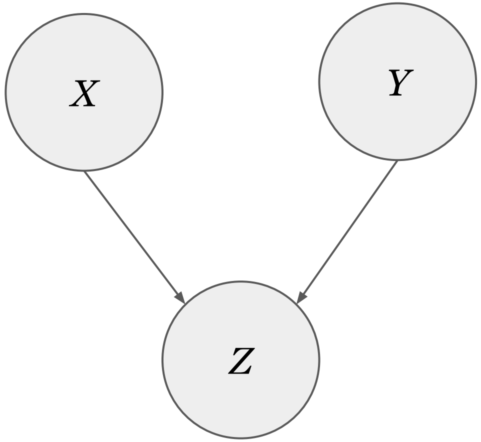


- Suppose we know $Z = 7$ (it's "given")

  then if $X = 1$, it must be $Y = 6$ …

  and if $X = 2$, it must be $Y = 5$ …

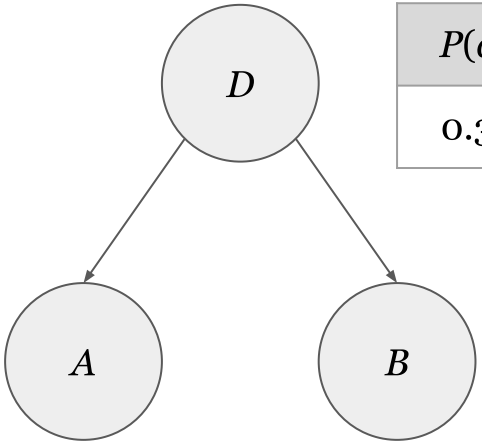
  $\Rightarrow$ Z creates a dependency between X and Y



Conditional probability tables

- In case of discrete variables, a **Conditional Probability Table** (**CPT**) is associated to each node
- If the BN in the figure represents the example of Slide 4, we can exploit the conditional independence of A and B given D to represent the full joint distribution with only **5** probabilities instead of the original 7 ⇒ **~30% less!**
- From these, we can derive all the probabilities of the joint distribution and therefore still answer any query
  - In the example, the tables do not need to specify $P(\neg d)$, $\mathbf{P}(\neg a|D)$, or $\mathbf{P}(\neg b|D)$ because they are implicit
    - $P(\neg d) = 1 - P(d)$
    - $\mathbf{P}(\neg a|D) = 1 - \mathbf{P}(a|D)$
    - $\mathbf{P}(\neg b|D) = 1 - \mathbf{P}(b|D)$

NOTE: lowercase letters are values, capital letters are variables, i.e. $\neg d$ means $D = false$



$P(d)$
0.31

D	$\mathbf{P}(a\|D)$
t	0.42
f	0.19

D	$\mathbf{P}(b\|D)$
t	0.55
f	0.29

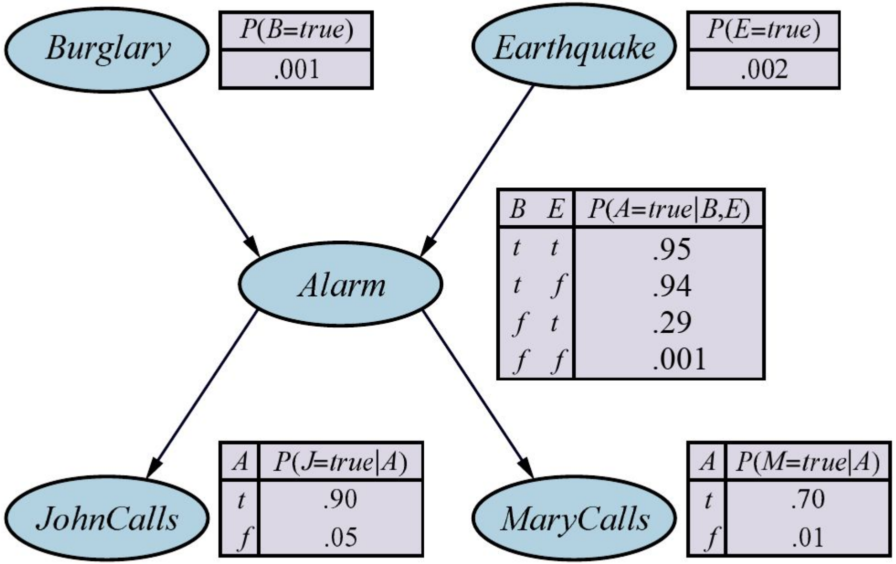
Advantages of product decomposition

- Besides **<u>storage space</u>**, the product decomposition represented by a Bayesian Network improves also the **<u>accuracy</u>** of the estimation
- Example
  - Estimate the probabilities of $\mathbf{P}(A, B, D)$ from a dataset with only 25 samples
    - since there are 8 possible triplets, we have 25/8 ≈ 3 samples, in average, for each case (in reality some might have many samples, some none…)
  - If we estimate $\mathbf{P}(A|D)$, $\mathbf{P}(B|D)$, and $\mathbf{P}(D)$, instead, it's more likely to have enough samples
    - 25/4 ≈ 6 samples in average for $\mathbf{P}(A|D)$
    - 25/4 ≈ 6 samples in average for $\mathbf{P}(B|D)$
    - 25/2 ≈ 12 samples in average for $\mathbf{P}(D)$

these "simpler" probabilities are based on **more samples ⇒ more accurate!**

Example: burglary alarm (from Russell & Norvig's book)

- A home security alarm system can be triggered by a burglar or an earthquake
- Neighbors *J*ohn and *M*ary should call the owner in case of alarm
- 5 CPTs (total size **10** instead of $2^5 – 1 = 31$)
- Example:



$P(B=true)$
.001

$P(E=true)$
.002

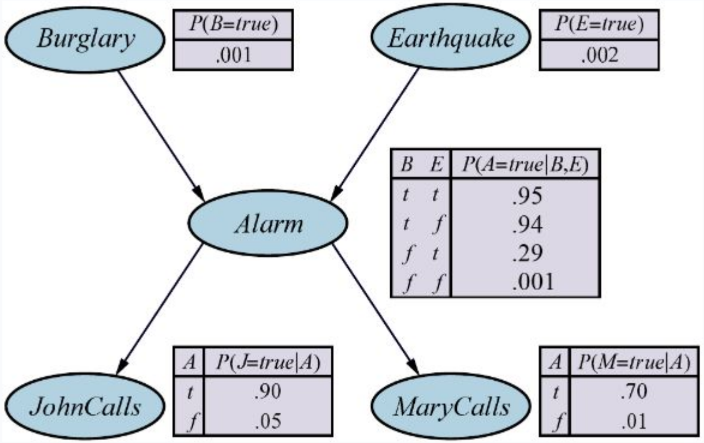
B	E	$P(A=true \mid B,E)$
t	t	.95
t	f	.94
f	t	.29
f	f	.001

A	$P(J=true \mid A)$
t	.90
f	.05

A	$P(M=true \mid A)$
t	.70
f	.01

$$P(j, m, a, \neg b, \neg e)$$
$$= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b)P(\neg e)$$
$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 \approx 0.00063$$

Burglary Alarm: P(j, ¬m, a, b, ¬e) = ?



0.00000118

0

0.00059102

0

0.00000051

0

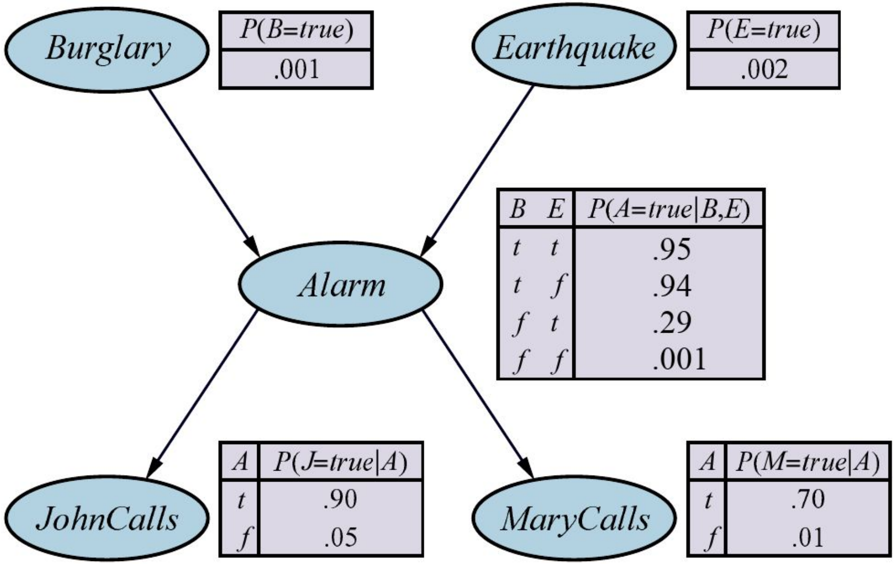
0.00025329

0

None of the above

0

Example: burglary alarm (from Russell & Norvig's book)

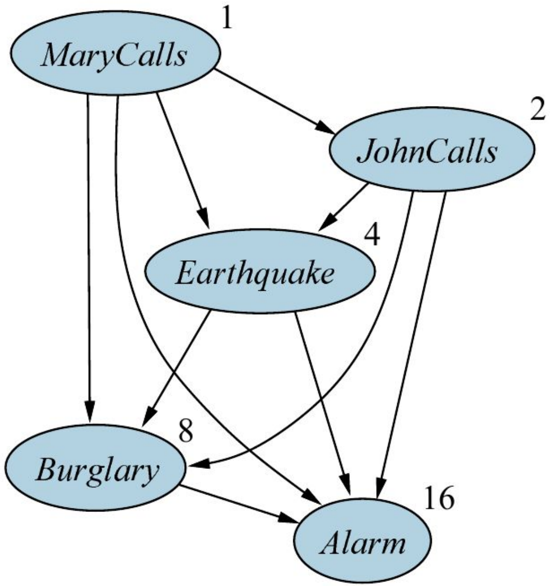


$$P(j, \neg m, a, b, \neg e)$$
$$= P(j|a)P(\neg m|a)P(a|b, \neg e)P(b)P(\neg e)$$
$$= 0.90 \times 0.30 \times 0.94 \times 0.001 \times 0.998 \approx 0.00025329$$

Network construction

- **Any node's ordering produces a correct Bayesian Network**

- However, **causal models** – i.e. where the $Parents(X)$ are the <u>causes</u> of X – are more efficient



- Example:
  $\mathbf{P}(A, B, E, J, M) = \mathbf{P}(A|B, E, J, M)\mathbf{P}(B|E, J, M)\mathbf{P}(E|J, M)\mathbf{P}(J|M)\mathbf{P}(M)$
  total size of CPTs = **31** >> 10

Inference by enumeration

- Let's start with a generic query to compute the probability of X given some evidence $\mathbf{e} = \{e_1, e_2, ..., e_n\}$

$$\mathbf{P}(X|\mathbf{e}) = \frac{\mathbf{P}(X,\mathbf{e})}{P(\mathbf{e})} = \frac{\mathbf{P}(X,\mathbf{e})}{\sum_X P(X,\mathbf{e})}$$

- The denominator is the sum over all possible values of X, therefore

$$\mathbf{P}(X|\mathbf{e}) = \alpha\mathbf{P}(X,\mathbf{e}) \qquad \alpha = \frac{1}{\sum_X P(X,\mathbf{e})}$$

where α is a **normalization** constant that guarantees the tot. probability is 1

Inference by enumeration

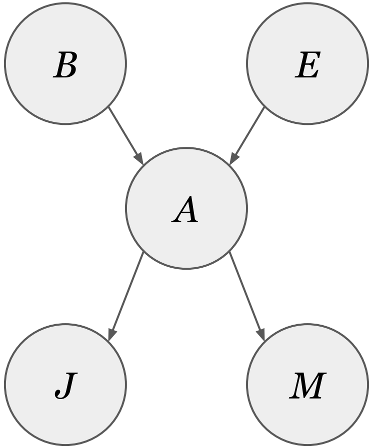
- With the joint distribution of a Bayesian Network, we can answer any query by marginalising (i.e. summing out) all the possible values $\mathbf{y} = \{y_1, y_2, \ldots, y_k\}$ of the **hidden** (unobserved) variables that are not part of the query itself

$$\boxed{\mathbf{P}(X|\mathbf{e}) = \alpha\mathbf{P}(X, \mathbf{e}) = \alpha\sum_{\mathbf{y}}\mathbf{P}(X, \mathbf{e}, \mathbf{y})}$$

- For example, the probability of burglary given that both neighbors called is

$$P(b|j, m) = \alpha P(b, j, m) = \alpha\sum_{e}\sum_{a}P(b, j, m, e, a)$$

Example



- Query

$$\mathbf{P}(B|j,m) = \ ?$$

- Solution

$$\begin{aligned}\mathbf{P}(B|j,m) &= \alpha\mathbf{P}(B,j,m)\\ &= \alpha\langle P(b,j,m), P(\neg b,j,m)\rangle\end{aligned}$$

- We can do <u>inference by enumeration</u> on the <u>product decomposition</u> of the BN to compute the probabilities $P(b, j, m)$ and $P(\neg b, j, m)$

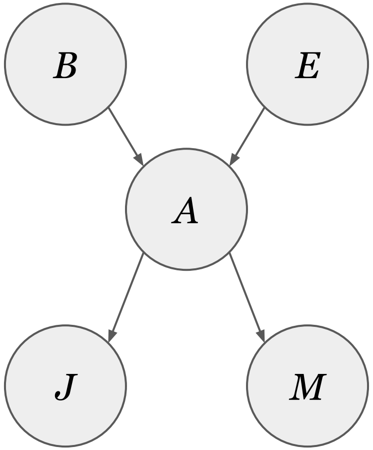
Example

not observed

$$P(b,j,m) = \sum_e \sum_a P(b,j,m,\boxed{e,a})$$

$$= \sum_e \sum_a P(b)P(e)P(a|b,e)P(j|a)P(m|a)$$

$$= P(b)\sum_e P(e)\sum_a P(a|b,e)P(j|a)P(m|a)$$



- We repeat the same for $\neg b$
- Finally, we normalize (with α) so that $\alpha\, P(b,j,m) + \alpha\, P(\neg b,j,m) =$

$$P(b|j,m) + P(\neg b|j,m) = 1$$

Example

$$P(b, j, m) = P(b)\sum_e P(e)\sum_a P(a|b,e)P(j|a)P(m|a)$$

$$= 0.001 \times [0.002 \times (0.95 \times 0.9 \times 0.7 + 0.05 \times 0.05 \times 0.01) +$$

$$0.998 \times (0.94 \times 0.9 \times 0.7 + 0.06 \times 0.05 \times 0.01)]$$

$$= 0.0005922$$

$$P(\neg b, j, m) = P(\neg b)\sum_e P(e)\sum_a P(a|\neg b,e)P(j|a)P(m|a)$$

$$= 0.999 \times [0.002 \times (0.29 \times 0.9 \times 0.7 + 0.71 \times 0.05 \times 0.01) +$$

$$0.998 \times (0.001 \times 0.9 \times 0.7 + 0.999 \times 0.05 \times 0.01)]$$

$$= 0.0014919$$

Example

- The distribution of the query is therefore

$$\mathbf{P}(B|j,m) = \alpha\langle 0.0005922, 0.0014919\rangle$$

$$\alpha = \frac{1}{0.0005922 + 0.0014919} \approx 479.823$$

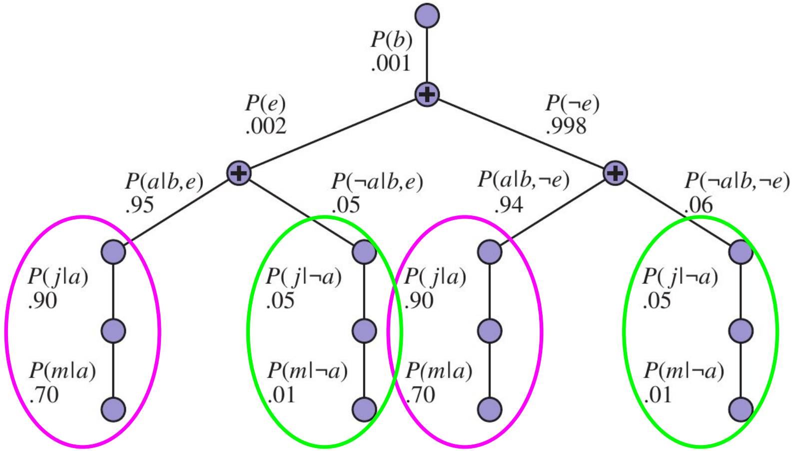
$$\mathbf{P}(B|j,m) = \langle 0.284, 0.716\rangle$$

Example

- Inference by enumeration is a depth-first, left-to-right recursive calculation process
  - e.g. $$P(b,j,m) = P(b)\sum_e P(e)\sum_a P(a|b,e)P(j|a)P(m|a)$$
    $$= 0.001 \times [0.002 \times (0.95 \times \underline{0.9 \times 0.7} + 0.05 \times \underline{0.05 \times 0.01}) + 0.998 \times (0.94 \times \underline{0.9 \times 0.7} + 0.06 \times \underline{0.05 \times 0.01})]$$

- However, some operations are repeated multiple times

- Possible improvement: save and reuse intermediate results (i.e. **variable elimination** algorithm)



Summary

- The **full joint probability distribution** specifies the probability of each complete assignment of values to random variables
- **Conditional independence** brought about by direct causal relationships in the domain allows the full joint distribution to be factored into smaller, conditional distributions
- **Bayesian networks** provide a natural representation for (causally induced) conditional independence
- **Topology + CPTs** = compact representation of joint distribution, generally easy for (non)experts to construct

Conclusions

- Suggested reading
  - Chapter 1, sec. 1.4 and 1.5.2 of Pearl et al.
  - Chapter 13, sec. 13.1-13.3 of Russell & Norvig

- Next lecture

  Bayesian Networks II

- Questions?